

Intermediate Elective

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Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(lubridate)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

Figure planning

Final visual goal: compare the abundances of the most common inverts found at ncos in 2025

Create dataframe

```

invert_2025 <- aq_ins |>
  clean_names() |>
  rename(date = date_on_vial) |>
  filter(year(date) == 2025) |>
  select(ostracod,
    copepod,
    hemiptera_corixidae_boatman,
    cladocera,
    nematode) |>
  pivot_longer(cols = ostracod:nematode,
    names_to = "taxon",
    values_to = "abundance") |>
  mutate(
    abundance = replace_na(abundance, 0),

```

```
    taxon = str_to_title(str_replace_all(taxon, "_", " "))  
  ) |>  
  group_by(taxon) |>  
  summarise(total_abundance = sum(abundance), .groups = "drop")
```